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DIP PBL Report

PBL Report on “Techniques in De-Blurring Image”

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Date: 13/11/2024

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PBL Report on "Techniques in De-Blurring Image" by 21CS105

Introduction

Images play a crucial role in various aspects of modern life, from scientific applications to social networking. However, image blurring poses a significant challenge, leading to degradation in image quality. Blurring can occur due to factors such as camera shake, motion, and noise, which can obscure important details and hinder accurate interpretation. The primary objective of this research is to explore and evaluate different de-blurring techniques that aim to restore high-quality images, thereby enhancing their utility across various fields.



Figure 1. (a) Average Blur (b) Motion Blur (c) Defocus Blur (d) Gaussian Blur.

Motivation for Research

The necessity for clear and detailed images is paramount in critical applications, including medical diagnostics, astronomical observations, and security surveillance. As imaging technologies advance, the demand for effective de-blurring techniques becomes increasingly vital. Existing methods often exhibit limitations, particularly in their ability to handle varying types of blur and noise. This research was motivated by the need to address these gaps and improve the quality of image restoration techniques.

Overview of Techniques Explored

The research delves into several types of blur, including:

- **Motion Blur:** Resulting from relative movement between the camera and the scene, often appearing as streaks in the direction of motion.
- **Gaussian Blur:** A widely used technique in graphics that applies a Gaussian function to reduce image noise and detail.
- **Average Blur:** A method that distributes pixel values across a defined radius R , effectively softening the image.

$$R = \sqrt{(g^2) + f^2}$$

Here, g was the direction of the horizontal size blurring, and f was the direction of the vertical blurring volume.

- **Out-of-Focus Blur:** Occurs when the camera fails to focus on the subject, resulting in a blurred image where some areas are sharp while others are not.

The deblurring models discussed in the research utilize mathematical formulations, particularly focusing on the convolution model and the point spread function (PSF), which describes how an image is distorted.

$$g = Hf + n$$

The g represents the blurred image and H is referred to as distortion factor, known as the point spread function (PSF).

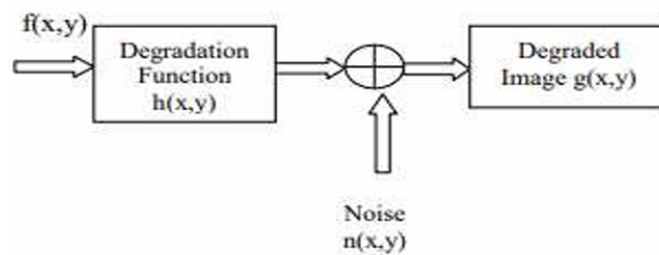


Figure 2. Deblurring image.

An original input is performed as a two-dimensional image $f(x, y)$ on system $h(x, y)$ and after adding noise $n(x, y)$, We obtain the degraded image $g(x, y)$. Digital image restoration may be seen as an operation in that, we attempt to approximate $f(x, y)$.

$$g(x, y) = (h(x, y) * f(x, y)) + n(x, y)$$

Methodology

The research evaluates several de-blurring techniques, including:

- **Deblurring Based on Edge Spread Function:** This approach leverages edge information from the blurred image to estimate the PSF. By analyzing the edges, the method reconstructs the original image effectively.
- **Neural Networks:** Utilizing neural networks allows for the approximation of complex operations required for image restoration. These networks can learn from input-output pairs to produce high-quality images.

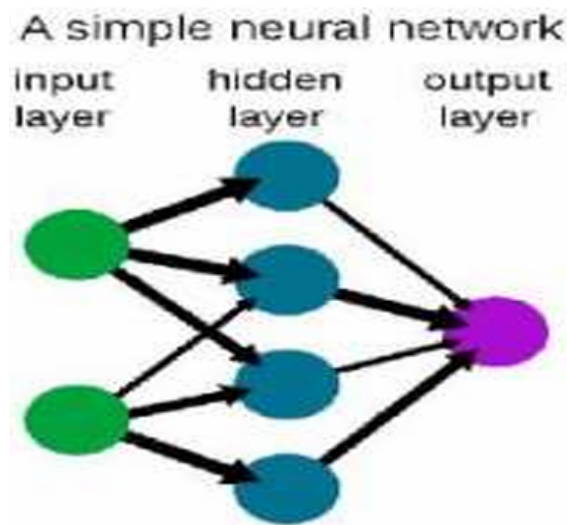


Figure 3. Neural Network Approach.

- **Blind Deconvolution:** This iterative method estimates both the PSF and the original image simultaneously. The process begins with initial estimates and refines them through repeated iterations until convergence is achieved.
- **Richardson-Lucy Algorithm:** A well-known iterative method for recovering a latent image from a blurred one. This algorithm is particularly effective when the PSF is known, but it requires careful consideration of the number of iterations to avoid artifacts.

Each of these techniques was implemented using appropriate software tools, as discussed in the research.

Benefits of the Research

The findings of this research highlight several benefits:

- **Improved Image Quality:** The techniques explored lead to significant enhancements in image clarity, enabling better feature extraction and analysis.

Method	Type of Blur	Performance	PSNR ratio
Weiner filter	Gaussian blur	Worst result	17.05
Regularized filter	Gaussian blur	Efficient	20.10
Lucy Richardson	Gaussian blur	Efficient	21.06
Handling outliers	Gaussian	Efficient	21.91
Hyperspectral (PCA)	Hyperspectral image blur	Efficient	22.34
MDF	Motion	Efficient	24.30
Motion density	Motion	Efficient	24.31
Blind deconvolution	Gaussian & motion	Efficient	26.78
Neural network	Gaussian & out-of- focus	Very Efficient	30.11
ASDS-AR	Gaussian	Very Efficient	31.20

- **Versatility Across Fields:** The applicability of these methods spans various domains, including medical imaging, astronomy, and security, demonstrating their broad utility.
- **Advancements in Algorithms:** The research contributes to the field by introducing novel algorithms and improving existing ones, particularly in the context of blind deconvolution.

Limitations and Challenges

Despite the advancements, the research acknowledges several limitations:

- **Computational Complexity:** Many of the algorithms, particularly blind deconvolution and Richardson-Lucy, are computationally intensive and time-consuming, which can be a drawback for real-time applications.
- **Noise Sensitivity:** Certain methods may struggle in the presence of high noise levels, which can adversely affect image quality and restoration accuracy.

- **Local Minima Problem:** The iterative nature of blind deconvolution can lead to convergence at local minima, which may not represent the best solution for image restoration.

Conclusion

This report presents a comprehensive review of state-of-the-art methods for image de-blurring. The exploration of blind deconvolution techniques shows that they generally outperform non-blind methods, particularly in challenging scenarios where the PSF is unknown. The Richardson-Lucy algorithm, while effective, requires careful management of iterations to mitigate ringing artifacts. Future research could focus on enhancing algorithm efficiency and improving resilience to noise, thereby further advancing the field of image restoration.

References

Muthana, R., & Alshareefi, A. N. (2020, May). Techniques in de-blurring image. In *Journal of physics: conference series* (Vol. 1530, No. 1, p. 012115). IOP Publishing.